Class 10

Chemistry

Volume – 2

Solutions

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3. REACTION MECHANISM

WORKSHEET – 1

LEVEL – I

1. B	2. C	3. B	4. B	5. D	6. A	7. D
8. A	9. C	10. D	11. D	12. C	13. B	

LEVEL – II

1. C	2. C	3. D	4. C	5. A	6. B	7. A
8. D	9. A	10. C	11. D	12. A	13. D	14. C
15. A	16. A	17. C	18. A	19. D	20. A	21. C
22. D	23. B	24. B				

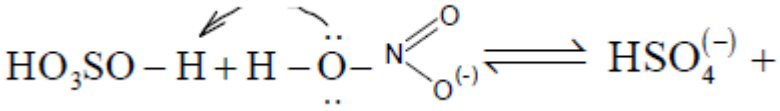
LEVEL – III

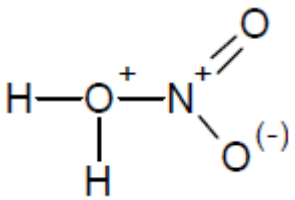
25. A, B, C, D	26. A, B, C	27. A	28. B	29. B	30. A

LEVEL – I

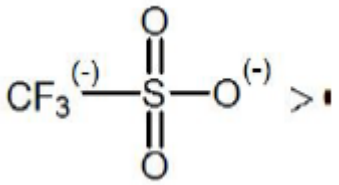
13. In basic medium epoxides give SN^2 mechanism in which nucleophile attacks less hindered carbon of the ring.

LEVEL – II

2. 



3. $3^0 > 2^0 > 1^0 > CH_3Cl$

4. 

$$CH_3 - SO_2 - O^{(-)} > I^- > CH_3O\,CO^{(-)}$$

7. Nitrobenzene is very poor nucleophile and alkyl carbocation is very poor electrophile.
8. NBS/ $h\gamma$ gives brominating reaction at allylic carbon, benzylic carbon and α-carbon of carbonyl compound.
9. Methyl halide always gives SN^2

10. The halogen attached sp^2 - hybridized carbon of the benzene ring due to resonance
11. α-halo ketone is most suitable substrate for SN^2 reaction α-halo ketone > $CH_3 - Cl > 1^0 > 2^0 > 3^0$
12. Reaction intermediate is benzyl carbocation presence of CH_3O - group at para position stabilized the RI.
13. Allyl carbocation is more stable than ally |> $2° > 1^0$
14. $SN^2: \gamma = K[\text{ alkyl halide }][^-OH]$

$\gamma_1 = K[R - X][OH] - - - -(1)$

$\gamma_2 = K[3R - X]\left[\frac{1}{2} \cdot OH^-\right] - - - -(2)$

$\frac{\gamma_1}{\gamma_2} = \frac{K[3(R-X)]\left[\frac{OH}{2}\right]}{K(R-x)(\bar{O}H]} = \frac{\gamma_2}{\gamma_1} = \frac{3}{2} = 1.5:$

$\Rightarrow \gamma_2 = \gamma_1 = 1.5 -$ times

18. For SN^1 reaction.
reactivity α stability of carbocation and reactivity α leaving power of the leaving group.
$[I^- > Br^- > Cl^- > F^-]$
19. $HCl + Axhy.\ ZnCl_2$ is called Lucas reagent. Reactivity order of alcohols with Lucas reagent is $3^0 > 2^0 > 1^0$
20. In unsymmetrical ethers, if both alkyl groups are primary then reaction is SN^2 and nucleophilic of the reagent attacks the alkyl group having smaller number of carbons.
21. Leaving power in decreasing order is as follows $-O - SO_2 - CF_3, OTs, Br, Cl$
22. It is a primary alkyl halide whose α-carbon is chiral
23. Reactivity α stability of free radical
24. In SN^2 reaction - reactivity order $-1^0 > 2^0 > 3^0$
25. $SN^2: CH_3 - X > 1^0 > 2^0 > 3^0$

WORKSHEET – 2

LEVEL – I

1. C	2. C	3. C	4. D	5. B	6. D	7. A
8. B	9. A	10. A				

LEVEL – II

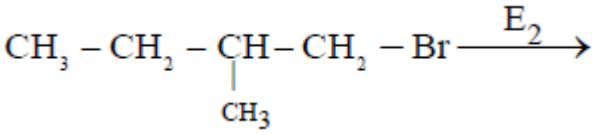
1. C	2. D	3. D	4. C	5. A	6. D	7. C
8. C	9. B	10. B	11. A	12. A	13. C	14. B
15. B	16. D	17. D	18. D	19. D	20. A	21. B

LEVEL – III

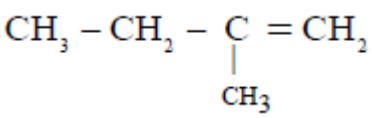
22. A, B, C	23. A, B, C	24. C	25. C	26. A	27. a-r, b-s, c-q, d-p
28. A, B, C	29. A	30. B	31. A, C	32. D	

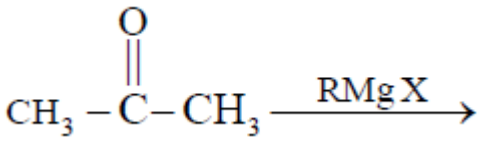
LEVEL – II

10. Alkyl fluoride gives Hafmann elimination [Poor leaving group]

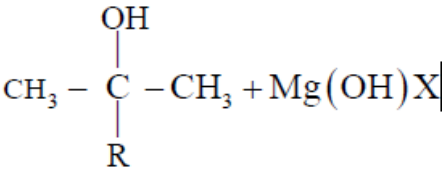


12.

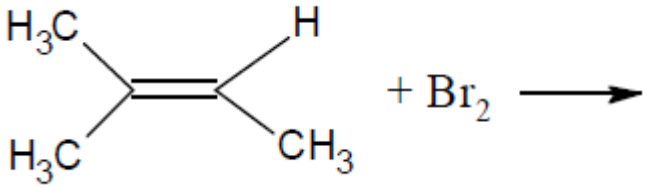




13. ketone gives teritiary alcohol.



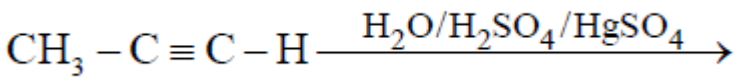
15. Heat of hydrogenation $\alpha \frac{1}{\text{Stability}}$

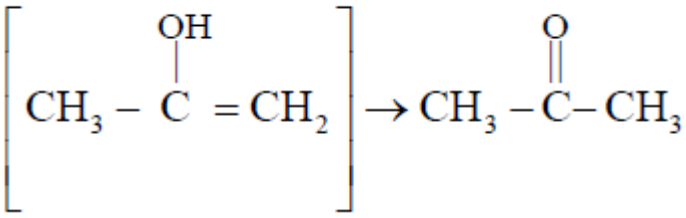


16. Optically inactive due to plane of symmetry.

17. 30-alkyl halide with strong base undergoes E2 reaction, to get more Elimination product.

19.





20. Alkene having following structure gives antimarkovnikov addition $H_2C = CH - Z$ where Z is strong - I group.

4. HYDROCARBONS

WORKSHEET – 1

LEVEL – I

1. A 2. D 3. C 4. B 5. C 6. D 7. B
8. C 9. D

LEVEL – II

1. D 2. D 3. B 4. A 5. D 6. 2 7. A
8.B 9. A 10. B 11. B 12.A 13. A, B 14. C, D
15. D 16. A 17. C 18. A 19. A

LEVEL – III

20. B, C 21. A 22.A 23. B 24. A 25. a-p, b-r, c-s, d-q
26. B, C 27. D 28.C 29. B 30. a-s, b-p, c-q, d-r
31. A, B, D 32. a-q, b-s, c-r, d-p

WORKSHEET – 2

LEVEL – I

1. D 2. B 3. C 4. D 5. C

LEVEL – II

1. A 2. B 3. C 4. B 5. C 6. B 7. C
8. C 9. B 10. B 11. A 12. C 13. A 14. A
15. B 16. B

LEVEL – III

17. A, B 18. A 19. A 20. B 21. C
22. a-s, b-r, c-q, d- 23. A 24. A 25. B 26. B

WORKSHEET – 3

LEVEL – I

1. A 2. C 3. B 4. C 5. A 6. D

LEVEL – II

1. B 2. A 3. C 4. C 5. B 6. B 7. C
8. B 9. A 10. A 11. D 12. B 13. C 14. B
15. B 16. B 17. D

LEVEL – III

18. A, D 19. A, B, C, D 20. B 21. B 22. B 23. A
24. A-P, B-Q, C-R, D-S 25. A, B, C, D 26. B 27. A 28. A
29. A